

MAINS MASTER PROGRAM (MMP) 2024

SCIENCE AND TECHNOLOGY-3

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2. BIOTECHNOLOGY AND HEALTH

1) VACCINE DEVELOPMENT

- Example Questions

- » Discuss the different ways in which vaccines are developed. [10 marks, 150 words]

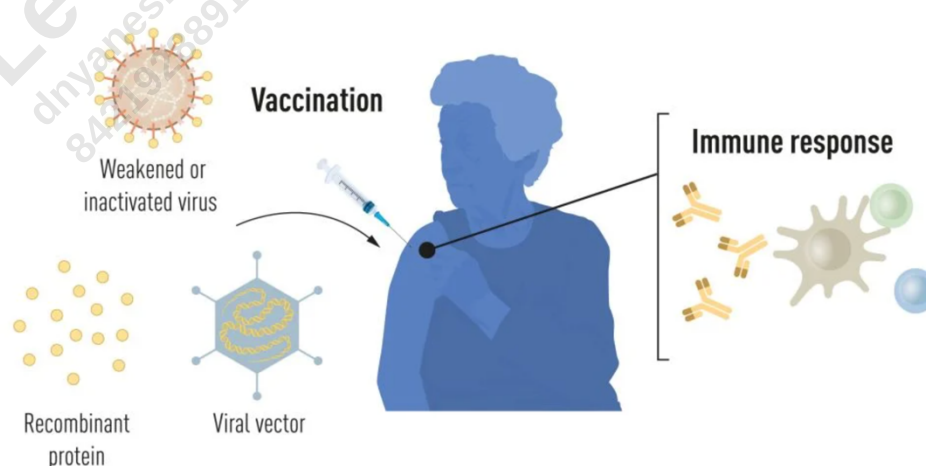
- Introduction:

- » A vaccine is a biological preparation that helps the body develop immunity against a particular disease. It typically contains a weakened or inactive form of disease-causing agent (like virus, bacteria etc.) or parts of it, which triggers an immune response without causing the disease.

- E.g. OPV given against Polio virus, Covishield against Corona virus

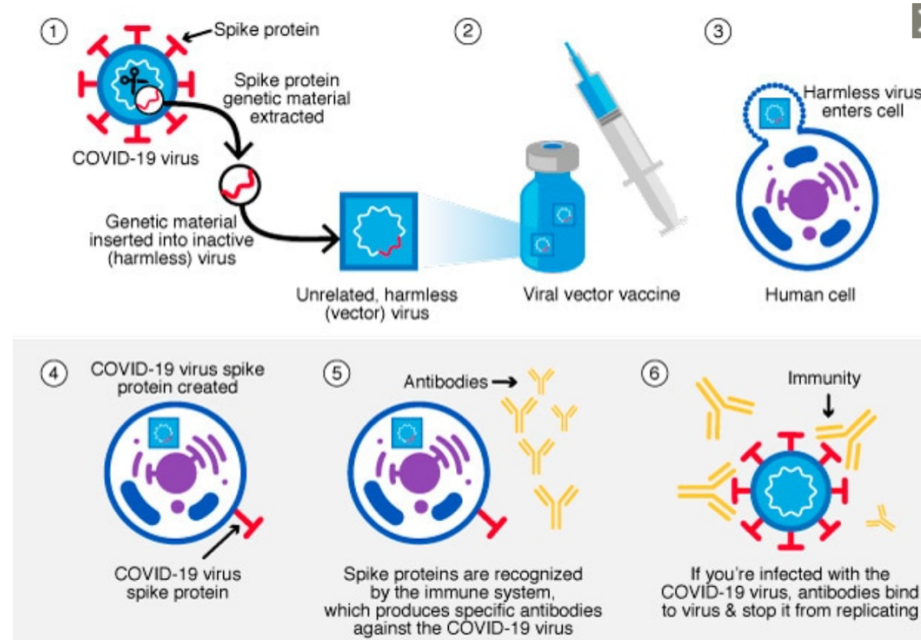
» Various Types of Vaccines:

- **Live Attenuated virus vaccines** such as the combined rubella-mumps-measles vaccines and the yellow fever virus vaccine, induce robust and long-lived antibody and T-cell mediated immunity.
 - » **Note:** For the development of Yellow fever vaccine, Max Theiler was awarded the Nobel Prize in Physiology or Medicine in 1951.
 - » These vaccines induce effective but transient immune responses, requiring repeated boosting.
 - » COVID-19 vaccine developed using this mechanism - Covaxin developed by Bharat biotech.
- **Viral Vector Vaccines:** It uses a safe virus (not harmful) which serves as a platform to produce target proteins to generate immune response.
 - » Such viral vector efficiently enter cells where the encoded antigen are produced by the bodies protein synthesis machinery.
 - The first example of a licensed viral vector vaccine was the Vesicular stomatitis virus -based vaccine against Ebola, approved in 2019, which was soon followed by an adenovirus based Ebola vaccine.



- **During COVID-19** various vaccines

- **Oxford-AstraZeneca** (ChAdOx1 nCoV-19) used adenovirus route.
 - **Covishield** used in India is a version of this.
- **Sputnik V Vaccine** also has gone adenovirus route.



- **Both the above methods** (live attenuated virus or viral vector vaccine) used cell culture-based manufacturing facilities which is **resource intensive**. Further they may also introduce diseases and is safer and stable than vaccine containing whole pathogens.
- Therefore, researchers have focused upon sub-unit vaccines that circumvent the need of large-scale cell cultures by delivering nucleic acid (DNA or mRNA) directly to vaccine recipients, exploiting the body's own capacity to produce proteins.
- **Subunit Vaccines:** (Protein subunit vaccines)
 - » Protein subunit vaccines **include only the parts of virus that best stimulate immune system**. These vaccines contain single protein components of the respective virus and are referred as subunit vaccine.
 - It includes Hepatitis B Vaccine (HBV) and Human papillomavirus (HPV) vaccine.
 - » **advantages:**
 - No risk of introducing the disease and is safer and stable than vaccine containing whole pathogens.
 - Suitable for immunocompromised individuals.
 - Well established tech
 - » **Disadvantage**
 - Relatively complex to manufacture (compared to other vaccines like RNA vaccines)
 - May require multiple doses.

- » **DNA and RNA subunit vaccines:**
- » **Advantages** of subunit vaccines (DNA or mRNA vaccines)
 - **Less Resource intensive** and thus easy to manufacture
 - **More flexibility** - Since the sequence can be easily changed to encode different antigens.
 - This also makes iterative testing of new candidate vaccines and generation of updated vaccines rapid and efficient.
- » **Initially DNA vaccine was thought to be more promising** but didn't translate into success. A likely reason for it was that injected DNA must cross two barriers, the plasma membrane and the nuclear membrane, to reach the cellular compartment where transcription takes place (DNA conversion to mRNA).
- » **Another advantage of mRNA vaccine:** Delivered nucleic acid can't integrate into the host genome. This is an additional safety aspect of this method.
 - **E.g of DNA vaccine** (developed for COVID-19):
 - **E.g for mRNA vaccine** (developed for COVID-19): Moderna COVID-19 (mRNA-1273) vaccine
- **Conjugate Vaccine**
 - » **Basics of Conjugate Vaccines:**
 - A conjugate vaccine is a type of subunit vaccine which combines a weak antigen with a strong antigen as a carrier so that the immune system has a stronger response to the weak antigen.
 - » **Soberana 02**, a conjugate COVID-19 vaccine developed in Cuba, was given emergency use authorization in Cuba and Iran.
- **Key Challenges**
 - » **Availability and affordability for all**
 - » **Ensuring complete Dose:** For e.g. in case of COVID-19 people have been careless about the 2nd dose of the vaccination.
 - » **People have been careless after the first vaccine and a large number of individuals**
 - » **Vaccine Hesitancy:** India has a history of hesitancy about receiving the polio vaccine due to various rumors.
 - Further, there is issues of limited health literacy, and a poor understanding of the protection vaccines provide
 - » **Vaccine Nationalism**
 - Countries like UK and USA had pre-booked more coronavirus vaccines than their population. This had the potential to delay the availability of vaccine in developing and under-developed countries
 - » **Vaccine Safety: Understanding short term and long-term impacts**
 - E.g. discontinuation of Moderna vaccine for younger age population.

- **Way Forward**
 - » **Proper Evaluation of vaccines** on technical parameters and suitability
 - **Immunity** of high degree; **broad scale** (against different variants); **durable** (at least five years of immunity); **Safe** (no side effects); **affordable**; **Programmatically suitable** (oral preference, no need of very cold storage etc.)
 - » **Fight Vaccine Hesitancy: Transparency, Public Education & Participation:**
 - Government will need a consistent, clear-eyed communication to bolster vaccine acceptance and community perception of the program. This will build trust and engage and inform all stakeholders especially experts.
 - Simultaneously, there is a need of **public education and information** through large-scale State and local-level networks where people are informed, sensitized, and their feedback taken.
 - Decision making should **be truly participative** where people who are taking the vaccine need to understand the program and be assured of vaccine's efficacy.
 - » **Fight vaccine inequity:**
 - **Free vaccination from government** and **paid vaccines from private sector** for people who want to "Pay and Get".
 - Bring vaccination centres to remote areas; ensuring second dose etc.
- **Prevent Vaccine Nationalism** - Promote international collaboration.
 - WHO in collaboration with other international bodies had launched a platform called ACT (or Access to COVID-19 Tools) Accelerator Program with an objective to accelerate the development, production and equitable access to COVID-19 vaccines.
- **Continue detailed long term study on vaccine safety and impact**
- **Focus on Eradication of Virus**
 - This will prevent new threats due to mutation or creation of reservoirs in animals.

2) MRNA VACCINES: 2023 NOBEL PRIZE IN PHYSIOLOGY OR MEDICINE

- The 2023 Nobel Prize in Physiology or Medicine has been awarded to Katalin Kariko (Hungary) and Drew Weissman (USA) for their discoveries concerning nucleoside base modifications that enabled the development of effective mRNA vaccine against COVID-19.
- **Background:**
 - » **Other methods of vaccine development** - Whole Virus -, protein-, and vector- based vaccines requires large scale cell culture. It is a **resource intensive** process and **limits the possibilities for rapid vaccine production** in response to outbreaks and pandemics. **mRNA based vaccines** solved these problems.
 - » **During the 1980s**, efficient methods of producing mRNA without cell culture were introduced, called in-vitro transcription. Ideas of using mRNA technologies for vaccine and therapeutic purposes also took off, but roadblocks lay ahead.
 - In vitro transcribed mRNA was considered unstable and challenging to deliver. It required development of sophisticated carrier lipid systems to encapsulate the mRNA.
 - This mRNA also gave rise to inflammatory reactions.

- These problems limited the enthusiasm for developing the mRNA technology for clinical purposes.
- **Contributions of the two scientists:**
 - They found that inflammatory response was almost abolished when base modification was included in the mRNA. Therefore, in 2015 they published that adjustments (modifications) to nucleosides, can keep the mRNA under the immune system's radar.
 - Later, they also showed that the delivery of mRNA generated with base modification markedly increased protein production compared to unmodified mRNA. This effect was due to the reduced activation of an enzyme that regulates protein production.
- **Development of Vaccines:**
 - After the above discoveries, interest in mRNA technology picked up. Vaccines for Zika and MERS-CoV were pursued.
 - **After the outbreak of COVID-19 pandemic,** two base-modified mRNA vaccines encoding the SARS-CoV-2 surface protein were developed at record speed.
- The impressive flexibility and speed with which mRNA vaccines can be developed pave the way for using the new platform also for vaccine against other infectious diseases. In the future, the technology may also be used to deliver therapeutic proteins and treat some cancer types.

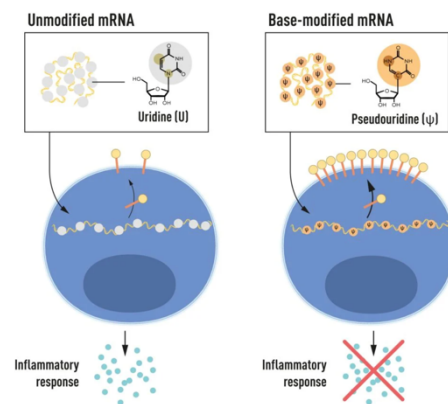


Figure 2. mRNA contains four different bases, abbreviated A, U, G, and C. The Nobel Laureates discovered that base-modified mRNA can be used to block activation of inflammatory reactions (secretion of signaling molecules) and increase protein production when mRNA is delivered to cells.
© The Nobel Committee for Physiology or Medicine. III. Matthias Karlén

3) APPLICATIONS OF BIOTECHNOLOGY

- The applications of biotechnology include therapeutics, diagnostics, genetically modified crops for agriculture, processed food, bioremediation, waste treatment, and energy production. This can be broadly divided in **four major industrial areas**
 - a. Agriculture and other food items (beer, wine)
 - b. Health care
 - c. Non-food (industrial) uses of crops and other Products (e.g. biodegradable plastics, vegetable oil, biofuels)
 - d. Environmental uses

A) AGRICULTURE AND FOOD PRODUCTION

ALCOHOL, CURD AND OTHER FERMENTED FOOD MATERIAL

GM ANIMALS

- In 1997, the first transgenic cow - Rosie, produced human protein-enriched milk (2.4 grams per litre).

GM CROPS

- Make crops more tolerant to anti-biotic stresses (cold, drought, salt, heat) etc.
 - E.g., GM Rubber developed by Rubber Research Institute of India
- Make plants **Pest Tolerant**.
 - Reduces reliance on chemical pesticides.
 - E.g. BT cotton, BT Brinjal (in Bangladesh)
- Help to **reduce post-harvest losses**
- Enhance the **nutritional value** of food, e.g., Golden Rice (Vitamin A enriched rice)
- Tailor-made plants to supply **alternative resources** to industries, in the form of starches, fuels, and pharmaceuticals

B) HEALTH CARE

- 1) **Diagnosis:** Molecular Diagnosis: RT-PCR and Rapid Anti-body tests
- 2) **Vaccines** (covered separately in details with health sector): For e.g., various vaccines for COVID-19 were developed with the help of biotechnology – mRNA vaccines, vaccines with attenuated viruses etc.
- 3) **Medicines produced by Biotechnology**
 - E.g., Genetically engineered insulin
- 4) **Gene Therapy** – It refers to process of introduction, removal, or change in the content of an individual's genetic material with the goal of treating the disease and a possibility of achieving long term cure.
 - E.g. CAR-T Cell therapy
- 5) **Potential of controlling mosquito borne diseases** – By introducing sterile mosquitoes (genetically formed)
- 6) **Personal Genomics** – Predictive, Precision, and stratified medicines.
- 7) **Transgenic Animals** – Rats etc. for testing of medicines, therapies etc.

C) BIOTECHNOLOGY AND ENVIRONMENT

- GM algae, crops etc. can provide more biomass for biofuel
- Fighting plant and animal diseases
- Waste Treatment / Bio-Remediation
- Biodiversity Conservation
 - E.g. -> De-extinction of species; **Colossal** is a new bioscience and genetics company, with the idea of bringing many extinct species back to life. Scientists at Harvard University in the USA would insert the Giant Woolly mammoth's (extinct 4,000 years ago) genes responsible for tiny ears, subcutaneous fat and hair length and Color into living elephant skin cells. Once they are successful in bringing these hybrids

back to life, Colossal will proceed with the ultimate goal of reviving the ancient extinct animals by producing more such hybrids.

- **Criticism:** Immoral; revival of these species may threaten the existing ecosystem and disturb the food chain which has evolved over the years; Rather than focusing on revival of long extinct species, biotechnology should focus on protecting the existing ones
- **Detecting Invasive Species:** environmental DNA based studies to detect invasive species like African catfish.

4) GETTING INTO THE DETAILS OF THE INDIVIDUAL APPLICATIONS

A) GM CROPS

- **Introduction:**
 - Crops whose DNA has been altered are known as GM crops. This genetic modification of crops can add or remove certain characteristics from the plant and thus can bring many advantages (already discussed)
 - **But** in India, the cultivation of GM crops has been a controversial issue right from the beginning. **BT Cotton** is the first and only GM crop which is being commercially cultivated in the country. Other crops like BT Brinjal and GM Mustard have still not been approved because of various environmental and health concerns.
 - However, **confined field trials** have been allowed for **20 GM crops**. These include Golden Rice and GM Rice as well.
 - But recent controversy surrounding BT Brinjal being grown illegally in Haryana and GM rice complaint by a French importer shows that some of these crops may have reached farmers illegally.

BT COTTON

- Specific BT Toxic gene (**cry1Ac**) were isolated from *Bacillus thuringiensis* and incorporated into several crop plants such as cotton. This produces proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), beetles, etc.
- It has been grown in India since 2002 and over the years have given increase productivity and area under crop cultivation. It has also led to decrease in insecticide which fought bollworms by 97%.
- But it has also raised concerns like increased water consumption, and emergence of pesticide resistant pests (e.g., pink bollworm), and increased use of insecticide for controlling pests like sucking pests.

BT BRINJAL

- Transgenic Brinjal created by inserting a **crystal protein gene (Cry1Ac)** from the soil bacterium *Bacillus thuringiensis* into the genome of various brinjal cultivar. It gives resistance against lepidopteron insects in particular the Brinjal fruit and shoot borer (BFSB), the most common pest which affects 30-50% of the Brinjal crops.
- The crop also cleared the GEAC's biosafety test in 2009. But government yielded to anti-GM activists and declared a moratorium in 2010 on the crop.
- But some cases of illegal BT Brinjal cultivation were observed in Haryana in 2019
- **Why are some groups are calling for allowing of BT Brinjal in India?**
 - It had cleared the GEAC's biosafety test in 2009.
 - **Increased benefit for farmers:** Popular all over the world. In Bangladesh studies shows zero borer infestation which has led to cut in input pesticide use by 60%.
 - When GM Crops are not officially available, farmers turn to **unapproved knock offs** that may not conform to accepted biosafety standards.
- **Why is BT Brinjal not allowed in India? Why is it opposed by various activists?**
 - There are fears that it may **impact India's plant biodiversity**. India has more than 3,000 natural varieties of Brinjal and activists are worried that if BT Brinjal is allowed in India all varieties will be contaminated.
 - Further, **cross pollination** may lead to **herbicide resistant super weeds** that can further threaten environment and biodiversity.
 - **Health Impact** is something that needs to be studied more.
 - **Not so obvious benefits:** A recent study from surveys of farmers indicate that 2/3rd of the farmers who moved to BT Brinjal have had a 'bad' or 'very bad' experience.

GM MUSTARD

- What is GM Mustard?
 - DMH-11 (Dhara Mustard Hybrid) is a genetically modified (GM) mustard Hybrid.
 - GM mustard is the country's first genetically modified food crop.
 - It was developed by a team of scientists led by former Vice Chancellor Deepak Pental, of DU at Center for Genetic Manipulation of Crop Plants (CGMCP), Delhi University by crossing Indian mustard cultivars with **juncea lines** of East European origin like 'Early Heera' and Donskaja.
- **Claim of higher yield:**
 - Claims around 30% more yield than the traditional varieties
- **What genetic modification was achieved and what are its benefit?**
 - **Barnase gene and Barster gene** from *Bacillus amyloliquefaciens*
 - Barnase impairs pollen production
 - Barster blocks the function of Barnase
 - Hybridization becomes possible:

- This method was used to develop DMH-11 by crossing a popular Indian mustard variety 'Varuna' (the barnese line) with an East European 'Early Heera-2' mutant (barstar).

▫ **Arguments for and against approval of GM mustard**

» **For**

- **Higher Production:** GM Cotton which has been used for over a decade has not resulted into any harmful affect. Country's cotton production has gone up by 2 and a half time since 2002 when GM cotton was first introduced.
- **Reducing Import Dependency:** India annually imports 3 million tonnes of Soybean oil (entirely GM) and another 0.4 million tonnes of rapeseed oil (partially GM). No NGO till date has blocked a single consignment of imported GM oil or even corn entering any port.
- **Saving Forex:** Will save a lot of foreign exchange and Balance of trade would improve.
- **Keeping India Scientifically relevant:** Approval was necessary for India to be scientifically relevant and to have better seeds to address threats from climate change.

» **Against**

- The main contention is that the GM mustard incorporates three alien genes - barnase, barstar, and bar - rendering it inherently unsafe for human and animal health.
 - **But** these genes have already been deployed in Canola, and we import it freely.
- Mustard is a food crop unlike cotton, so both should not be compared
 - But cotton seed also yield oil and oilcake.
- All health effects not properly known yet
- Environmental damages should be studied properly first.
 - GM Cotton has led to higher requirement of water and pesticides not only increasing the cost but also harming the environment
- Yield claims have been challenged by many organizations

» **GEAC Approval (Oct 2022)**

- In Oct 2022, GEAC approved environmental release of two varieties of genetically engineered mustard, so that it can be used for developing new parental lines and hybrids under the supervision of ICAR. The environmental release of DMH-11 will allow for its seed production and testing as per existing ICAR guidelines and other extant rules/ regulations prior to commercial release. The field demonstration studies on the effect of GE mustard on honeybees and other pollinators was also allowed to be conducted.

GM RUBBER – DEVELOPED BY KERALA BASED - RUBBER RESEARCH INSTITUTE OF INDIA

- Rubber Research Institute of India have developed a plant tailored for the climatic conditions in the Northeast.

- Rubber board research farm at Sarutari on the outskirts of Guwhati now sports world's first GM rubber plant, tailored for climatic condition in the north-east.
- **Genetic Modification:** The GM rubber has additional copies of the gene MnSOD, or manganese-containing superoxide dismutase, inserted in the plant, which is **expected to tide over the severe cold conditions during winter** – a major factor affecting the growth of young rubber plants in the region

INCREASING THE NUTRIENT CONTENT – GOLDEN RICE

- Golden rice (developed by IRRI) is variety of rice produced through genetic engineering to biosynthesize beta-carotene. Beta-carotene is a nutrient similar to what is found in orange-colored fruits and vegetables and is converted into Vitamin-A as needed by the body.
- Thus, golden rice can help south and south-east Asian countries, where two-thirds or more of daily calorific intake is obtained from rice. Research has indicated that the golden rice can provide upto 50% of the daily requirement of an adult for vitamin A.

D) WAY FORWARD

- **Detailed scientific analysis of GM Mustard, GM Brinjal, HTBt Cotton:** Considering the advantages of GM crops, it is crucial that government through GEAC goes through detailed scientific analysis of both GM Mustard and GM Brinjal once again. We need **more rigorous and sensitive studies** than those conducted so far. This should be done on priority in a fast track manner.
- **Study international Examples** understand the impact of crops like BT Brinjal before reaching any conclusion.
- **Strengthen Regulatory Framework:** There should be steps taken to ensure that illegal/accidental uses of GM Mustard/Brinjal/Rice/HBT Cotton doesn't take place.
- For this there has to be steps taken on multiple front – from farmers' education/awareness to action against vendors supplying the illegal GM Seeds. This would require a **thorough probe** and an **efficient regulatory mechanism** in future.
- Finally, there should be steps taken to **enhance the labelling infrastructure** so that GM food items can be properly labelled, and citizens can practice their right to choose by accepting or rejecting such food crops.

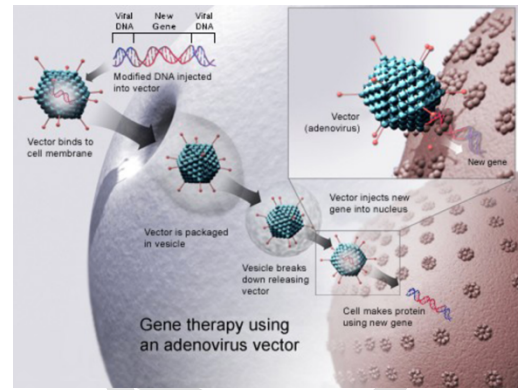
5) GENE THERAPY

» Example Questions

- Gene therapy is gaining popularity in developed countries to treat various types of rare genetic disorders. Describe briefly what gene therapy is and what advantages it has over other treatments? How are National Guidelines for Gene Therapy Product development and Clinical Trials, 2019 going to contribute to the gene therapy development in the country. [250 words, 15 marks]

» Introduction

- If a person is born with a hereditary disease, can a corrective therapy be taken for such disease? Gene therapy is an attempt to do this.
- **Gene therapy** refers to the process of introduction, removal or change in the content of an individual's genetic material with the goal of treating the disease and a possibility of achieving long term cure.
- **Gene Therapy Products (GTPs)** include the mechanisms to deliver nucleic acid components by various means for therapeutic benefit to patients. They include entities that are used for things like gene augmentation, gene editing, gene silencing, synthetic or chimeric gene augmentation etc.



» Advantages of promoting gene therapy

- **Permanent treatment for genetic diseases:** the therapeutic approach earns the advantage of being long lasting, sometimes permanent result.
- **High burden of rare genetic diseases in India:** Around 7 core of India's population suffers from rare genetic diseases. Gene therapy can prove to be a turning point in treatment of such genetic diseases.
- **Worldwide market** of upto \$250 billion by 2025; continuous scope of **medical tourism**.
- In USA and Europe, the approval of several gene therapy products in USA and EU have paved the way for development of therapies for a variety of previously untreatable disorders. The guidelines should help in development and fast tracking of approval of such GTPs in India too.

» Concerns/Limitations

- Promotion of development of gene therapy also brings along with it unique technical risks and ethical challenges.
- **Technical Challenges**
 - The gene therapy may be associated with **unwanted immune system reactions**. For e.g. when vectors (viruses) are attacked by the immune system of the body.
 - Current gene therapy mechanisms can sometimes **target the wrong cells**.
 - **The delivery viruses may mutate** and become harmful.
- **Ethical Challenges**
 - For e.g. creation of GM babies using germline gene editing by a Chinese scientist attracted global criticism and fuelled debate on ethical concerns regarding applications of gene therapy technologies.
 - **Playing god debate**

- **Conclusion:**

- Gene therapy will create more choices for patients who currently live in challenging, or with incurable diseases. The 2019 guidelines for gene therapy in India is the first step towards development of GTP in the country. India needs to increase investment in the sector, R&D support avenues, and education and awareness amongst the clinical community to make India a leading player in gene therapy.

6) CAR-T CELL THERAPY

» **Practice Questions:**

- Explain the mechanisms of CAR-T Cell therapy? Discuss the key advantages of NexCAR19 over other foreign CAR-T cell therapies [15 marks, 250 words]

» **Background: How Cancer has been treated before CAR T-Cell Therapy:**

- **Surgery** (removing the cancer cells)
- **Radiotherapy** (delivering ionizing radiation to the tumour)
- **Systematic Therapy** (administering medicines that act on tumour)
 - The earliest form of systematic therapy was **chemotherapy**. It preferentially acts on cancer cells because of the latter's rapid, unregulated growth and poor healing mechanisms. These drugs have modest response rate and significant side effects as they effect numerous cell types in the body.
 - The next stage in its evolution was **targeted agents** a.k.a. **immunotherapy**: The drugs bind to specific target on the cancer or in the immune cells that help the tumour grow or spread. This method often has less side effects as the impact on non-tumour cells is limited. However, it is effective only against tumours that express these targets.

» **CAR-T Cell Therapy** has emerged as a new development in this front.

- It is a revolutionary therapy that modifies immune cells, specifically T-Cells, by turning them into potent cancer fighters known as CAR-T Cells.
- **How it works?**
 - In CAR T-cell therapy, the patient's blood is drawn to harvest T-cells – immune cells that play a major role in destroying tumour cells.
 - **Researchers modify these cells** in the laboratory so that they express specific proteins on their surface, known as **chimeric antigen receptors** (CAR): they have an **affinity for proteins on the surface of tumour cells**. This modification in the cellular structure allows CAR T-cells to effectively bind to the tumour and destroy it.
 - These modified cells are then infused back into the patient's blood stream after conditioning them to multiply more effectively.
 - The cells are even more specific than targeted agents and directly activate the patient's immune system against cancer, making the treatment more clinically effective. This is why they are called '**living drugs**'.
- **Advantages of CAR-T Cell therapy over other Cancer fighting methods:**

- It is very accurate and only targets cancer cells.
- It makes the treatment easier with onetime therapy (unlike several sessions of chemotherapy)
- It can also fight non-responsive cancer patients.
- It is designed to cure and provide lifelong benefits.

» **CAR T-Cell Therapy in India:**

- The first major clinical trial showing they were effective was published almost a decade ago. The first indigenously developed therapy in India was successfully performed only in 2021.

» **In Oct 2023, the Central Drugs Standard Control Organization (CDSCO) granted market authorization for NexCAR19, India's first indigenously developed CAR-T cell therapy, to ImmunoAct, a company incubated by IIT Bombay. This paves the way for commercial launch of this therapy in India.**

- It is designed to target cancer cells that carry the CD19 protein. This protein acts like a flag on cancer cells, which allows CAR-T cells to recognize and attach themselves to the cancer cells and start process of elimination.
- **Who can get the NexCAR19 therapy?**
 - The therapy is for people with B-Cell lymphomas who didn't respond to standard treatments like chemotherapy, leading to relapse or reoccurrence of the cancer.
- **Significance:**
 - India is one of the first developing country to have its own Car-T therapy. Even some developed nations don't have their own CAR-T therapies and they import from USA or Europe.
 - This reduces the cost of treatment to about 1/10th of the cost abroad and has the potential of boosting medical tourism in India. It costs around Rs 3.3 crores abroad while in India it will cost somewhere between 30-40 lakh rupee.
 - Lab and animal studies have shown that **NexCAR19** lead to significantly lower drug-related toxicities. For e.g., it causes minimal damage to neurons and the central nervous system, a condition known as neurotoxicity. The therapy also leads to minimal Cytokine Storm Syndrome (CRS), which is characterized by inflammation and hyperinflammation in the body due to the death of a significant number of tumour cells, as CAR-T cells are designed to target and eliminate cancer cells.

7) GENE EDITED CROPS

- **Gene-Edited crops** are trans-gene free and contain no foreign genes. Gene editing tools are used to generate changes to native genetic material to yield beneficial outcomes.

- **Regulatory framework** for gene editing are nascent and generally less prohibitive relative to GMOs. Legislations regarding gene editing is emerging globally and trending towards allowing gene-edited products to pass from research to production with relative ease, compared to GMOs.
- **Gene Editing Tools** can be used to produce GMOs. In this case, novel configurations of genetic material can be precisely inserted into the genomes of organisms by using gene editing machinery.
- Gene-Modified Crops contains foreign genes. They leverage the genetics of other organisms to improve desired traits. (e.g. BT-Cotton, DMH-11, Golden Rice)

A) GENE EDITED MUSTARD – LESS PUNGENT, MORE USEFUL

- **Understanding the Problem:**
 - Mustard/Rapeseed is one of the most significant oilseed grown in India. But, **mustard seeds** have very high levels of **glucosinolates**, a group of sulphur and nitrogen-containing compounds contributing to the characteristic **pungency** of their oil and meal. This limits the acceptance of mustard oil by many users specially those who are used to less strong odour and flavour. The problem is even more in case of meal (the residual cake after extraction of oil from the seeds). Rapeseed meal is **unpalatable to poultry and pigs**, while having to be mixed with fodder grass and water for giving to cattle and buffaloes. Moreover, high glucosinolates are also known to cause goitre (swelling of neck) and internal organ abnormalities in livestock.
- **Efforts to improve the quality of Mustard** and the problem:
 - **Scientists have bred low glucosinolates variety of mustard, but large scale cultivation couldn't take place.** This is because reducing glucosinolates increases the vulnerability of crops to pests and diseases.
 - So, what is needed is to reduce the glucosinolates level in seeds, without lowering the levels in rest of the plant.
- **The Gene Editing Breakthrough:**
 - **Glucosinolates** are synthesized in the leaves and pod walls of mustard plants. They are transported and **accumulated in seeds through the action of glucosinolates transporter or GTR genes.** There are 12 such genes under two distinct classes of GTR1 and GTR2 with six copies each.
 - **Scientists edited 10 out of 12 GTR genes** in 'Varuna', a high yielding variety of Indian mustard. They used CRISPR/Cas9 tool for this. This editing made changes in the encoded proteins which were responsible for transport of the glucosinolates to the seeds.
 - **Result:**
 - **GTR Edited Low-seed high-leaf glucosinolate:** Seeds of GE Varuna mustard variety has glucosinolates content well below the **30 ppm canola quality.** Other parts of the plant, especially the leaves and pod walls enclosing the seeds, has significantly higher glucosinolate accumulation.

- **Resistance against pest is intact:** The edited variety continues to display defence against virulent fungal pathogen *Sclerotinia sclerotiorum* and the insect pest *Spodoptera litura*.
- **GE crops are subjected to less stringent “environmental release” regulation in India.**
 - For **GM Crops**, clearance has to come from Genetic Engineering Appraisal Committee and MoEF&CC (final nod). But, for GE crops requirement is less stringent.
 - In **March 2022**, an office memorandum from the MoEF&CC exempted GE Plants “free of exogenous introduced DNA” from the requirement of GEAC approval for open field trials leading to commercial release. Such clearance is now necessary only at the level of Institutional Bio-Safety Committee (IBSC), comprising scientists from the institutions engaged in the GE Crop development and from the DBT.
- **This work will increase the acceptability of mustard oil both within country and in the export market.**

Conclusion:

GM Hybrid Mustard (DMH-11) and the new GE low-seed and high-leaf glucosinolate lines are major plant breeding advancements – from Indian scientists. It can go some way towards bringing down the dependence on imported vegetable oil.

8) ANTI-MICROBIAL RESISTANCE

- **Example Questions:**
 - » Antimicrobial resistance is a multi-driven problem and only a multi-pronged approach can be helpful in tackling the scourge. Elaborate [10 marks, 150 words]
- **Introduction:**
 - » Antibiotic resistance occurs when the microbe has developed immunity against the antibiotic that was used to kill those microbes. In this situation, the bacteria become "resistant" and continue to multiply in the presence of therapeutic levels of antibiotic.
- Why do bacteria become resistant to antibiotic
 - » **Natural Phenomena: Evolution** - Selective pressure for the survival of resistant strains of bacteria.
 - » **Human Action:** Current higher levels of antibiotic resistant bacteria are attributed to the overuse and abuse of antibiotics.
- **Why Anti-biotic resistance is more prevalent in India: Key Factors**
 - » **India is the largest consumer of anti-microbials globally** and the use of **last resort anti-microbials** like cephalosporins is soaring.
 - **Easy availability and overuse** of anti-biotics is the most important factor: Over the Counter Availability; Irrational Use; over-prescription by doctors

- For e.g. Children often receive multiple courses of antibiotic every year since the viral infections are recurrent. This makes them more vulnerable to anti-microbial resistance.
 - » **Poor Health Sector** -> improper treatment -> Development of anti-biotic resistance
 - Further, exposure to subtherapeutic levels of anti-microbials or non-adherence to prescribed medications has also been cited as a driver of AMR
 - E.g.: in case of TB
 - =
 - » Increasing and completely unregulated use of antibiotic in Agriculture, live stocks and Poultry sector.
 - For instance, Colistin is extensively used in veterinary practices as a growth promoter. This leads to generation of colistin-resistant bacteria in poultry and freshwater fish.
 - » **Poor Sanitation conditions** -> More diseases -> More use of medicines -> More AMR development
 - » Unchecked discharge of effluents by the pharmaceutical industries
- **Impact of increasing anti-microbial resistance**
- **Damage to Public Health:**
 - » In 2019, drug-resistant superbugs killed about 1.27 million people globally - a toll more than HIV/AIDs or malaria - and according to the UN estimates, the number could reach 10 million by 2050.
 - » Demands complicated treatment pattern, with longer stay in hospitals -> increase in cost of treatment.
 - » Stronger antibiotics which are used after the first line of drugs fail generally have toxic side effects
 - » Resistance also emerging for second line of drugs (e.g. XDR-TB emerging)
 - » Without functional anti-microbials to treat bacterial and fungal infections, even the most common surgical procedures, as well as cancer chemotherapy, will become fraught with the risk of untreatable infections.
 - » All this is compounded by the fact that no new class of anti-biotics have made it to the market in the last three decades, largely on account of inadequate incentives for their development and production.
 - **Economic damages** due to AMR can be equivalent to what 2008-09 economic shocks resulted into: UN Report
 - **Environmental Damages**
 - » Extensive amount of anti-biotics led to development of AMR in some micro-organisms. It impacts the microbial biodiversity and thus the environmental balance needed.
- **Steps that government has taken and Steps that we further need to take**
- **National Policy** for Containment of Antimicrobial Resistance, 2011
 - Guidelines for appropriate antibiotic usage which have revised Schedule H drugs to make over-the-counter availability of certain antibiotics nearly impossible
 - Programs such as Red Line Campaign
 - Sanitation campaigns such as Swachh Bharat Mission etc.
 - National Surveillance system for AMR (April 2017)

- **National Action Plan on Antimicrobial Resistance (April 2017):** Focused on enhancing awareness, strengthening surveillance, improving rational use, promoting research and supporting neighboring countries

- **What more could be done**

- **Strengthen healthcare services** -> early detection, high quality medicines, complete treatment, infection prevention in health care facilities and farm.
- **Proper Implementation** of National AMR resistance action plan should get high priority, towards tackling drug resistance.
 - » Efficient utilization -> Following WHO's '**Access, Watch and Reserve**' strategy.
 - » Strict implementation of various drugs control regulation and increasing the fine for over-the-counter sale of drugs.
 - » Import and Export policies of food and feed should strictly regulate the anti-biotic use.
- **Proper regulation of livestock sector** to reduce the use of anti-biotics there.
 - » **Improve biosecurity** and ensure that harmful pathogenic organisms are not present at the farm.
 - » Promote vaccination over drugs
 - » **Developing and Using Alternatives**
 - **Botanical products with anti-microbial properties:** Extracts from turmeric, ginger, pepper and garlic are effective anti-microbial and can be added to the feed to control bacteria.
 - Use of **Probiotics and Prebiotics**
 - Using Seaweed extracts for improvement of immunity and additional physiological performance among aquaculture animals.
 - **Using enzymes** instead of antibiotics as growth enhancers. For e.g. enzymes like proteases, amylases, cellulase, esterase, lipase etc. are intended to enhance the availability of nutrients and help nutrient absorption in the digestive system.
 - **Phage therapy** (i.e. using Bacteriophages in aquaculture)
 - **Irradiation of food crops**
 - » DAHD should develop standard treatment guidelines to reduce misuse of anti-biotics.
 - » Bring a law to regulate manufacturing and sale of poultry feed laced with anti-biotics.
- **Regulating pharmaceutical industry effluents** -> strengthening BWM rules and improving its strict implementation.
- **Tackling AMR in Environment**
 - » Come up with a technical guidance to contain AMR from waste and environment.
 - » Shift to safer manufacturing practices for pharma to ensure reduced wastage and discharge in environment.
- **More Research** in the field of Anti-biotic resistance and coming up with safe mechanism to treat these AMR diseases
 - » **A multi-sectoral \$1 billion AMR Action Fund** was launched in 2020 to support the development of new anti-biotics. Similar steps to allocate more resources for AMR research would be required.
- **International Collaboration** should increase.
 - » All UN member states should phase out the use of anti-microbial on the WHO's highest priority list as growth promotion agents.

- » Trade of anti-biotics must be regularized for therapeutic purpose only with strict legalized medical prescription and supervision.
- **Increased focus on awareness** generation among common people against the need of excessive use of anti-biotics.
- **Conclusion**
 - The world can't contain anti-microbial resistance unless stakeholders from all sectors such as human and animal health, environment, crops, food and drug come forward to act. **One-Health action** is must to slow down AMR chronic.

9) GENERIC MEDICINES

- **Example Questions**
 - » "Generic medicines can play a key role in making India's health sector affordable". Discuss. [15 marks, 200 words]
- **What is a generic drug?**
 - » Generic drug is a **low-cost version of pharmaceutical drug** that is equivalent to a brand-name product in dosage, strength, route of administration, quality, performance and intended use.
 - » They usually enter market after patent protection of the original drug expires.
- **Note:** Broadly Medicines can be of three types:
 - » **Branded:** These are still on patent
 - » **Branded Generic:** Off Patent and Generic, but nonetheless produced by a reputed company, with a brand.
 - » **Generic:** Off Patent, and unbranded.
- **Government's attempt in promoting generic drugs**
 - » **National Medical Council (NMC)** had directed all doctors to prescribe generic names and not branded names
 - But this order was withdrawn after protest from Indian Medical Association.
 - » Making it mandatory for all chemists to display generic medicines prominently
 - » **Pradhan Mantri Bhartiya Janaushadhi Pariyojna (PMBJP)** was launched by Department of Pharmaceuticals, Ministry of Chemical and Fertilizers, Government of India as a direct market intervention scheme in 2008.
 - As of Aug 2023, **9,500 Jan Aushadhi Kendras** are functional across the country.
 - The government has also launched a mobile application 'Janaushadhi Sugam' that provides information to public about location of Kendras, help them search the Janaushadhi medicines, and compare the MRP of Generic vs Branded medicines.
- **Advantages**
 - » **Affordable healthcare:** Generic medicines are cheaper due to low marketing cost and no commission to doctors.
 - » **Breaks the doctor-pharma nexus:** The existing nexus leads to prescription of only the brand of companies

- **Reduce unnecessary prescription:** This is resorted to by doctor if pharma companies are paying them
 - » **Promotes domestic pharma companies:** India is a world leader in generic drug manufacturing.
 - » **Difficult for quacks to function** as non-qualified people may find it difficult to know generic names
- **Limitations**
 1. **Quality concerns**
 - Concerns about lack of uniformity in the quality of generic versions.
 - Studies have shown that many generic versions don't work.
 - Doctors trust and prefer well-established brands.
 2. **Erode doctor-patient relationship**
 - As by prescribing generic drugs, doctor refuse to take responsibility for clinical outcomes.
 3. **Low profit margins for retailers:** Retails pharmacists, in turn, have little incentive in stocking and selling low price generic medicines since they have lower profit margins.
 4. **Shortage:** Though, there are more than 9,000 Jan Aushadhi Kendras, the reach of traditional medical stores is very high and they primarily deal in non-generic versions.
 5. **Difficult for common person to understand**, especially the multiple salt names in a FDC.
 6. **May discourage big pharma companies to launch their new medicines in India**
- **Way forward**
 - » **Improved Quality** through improved regulation of pharma sector.
 - Human Resource issues like shortage of drug quality inspectors has to be tackled in a fast-track manner.
 - » **Increasing Production and Availability:** Increasing the penetration of Jan Aushadhi Kendras.
 - » **Allow pharmacist to substitute for alternative:** Laws for enabling substitution of generic and branded equivalents by pharmacists need to be introduced.
 - » **Prohibit Branding for out of patent drugs**
 - » **Increased awareness** on generics needed among consumers, pharmacists
 - » **Short names for generic FDCs** (officially approved trade names) will also ensure that doctors don't write out the generic names of their multiple constituents.
 - » **Improving government health facilities** would contribute to more coverage of government hospitals in overall health coverage and government hospitals can promote generic better.
- **Conclusion**
 - » Overall, generic prescribing is a good move in the right direction and will have several positive ramifications for healthcare in India. However, like everything, successful implementation would require a series of enabling steps at different levels from production to prescription and from Quantity to Quality.

10) HOOCH TRAGEDY

- **Practice Questions:**
 - » What is Hooch tragedy? Discuss the key factors responsible for persistence of illicit liquor production and consumption in India [10 marks, 150 words]
- **Introduction:**
 - » **(Current Affairs)** More than 60 people died in Kallakurichi hooch tragedy in Tamil Nadu in June 2024. Earlier in 2023, the state of TN saw death of 20 people and in March 2024, around 20 people had died due to Hooch in Sangrur Punjab.
 - » **(Definition)** Hooch tragedy refers to the deaths and disabilities caused by consumption of poisonous spurious liquor.
- **Why does spurious liquor become poisonous?**
 - » **Excess Methanol** can be toxic as methanol metabolizes into formic acid.
 - » **Improper preparation also leads to increase in concentration of methanol:**
 - The fermented mixture which is distilled also contain methanol. During distillation, both ethanol and methanol are concentrated. Thus, if done incorrectly, distillation can lead to an end product which has high concentration of toxic methanol.
 - » **Excess methanol** is toxic as it metabolizes into formic acid.
- **Why is methanol used instead of alcohol?**
 - » **Easier availability:** In industry it is used as antifreeze, solvent, fuel and ethanol denaturant.
 - » **Taste and Appearance:** Very similar to alcohol.
- **Why do people go for this kind of drink**
 - » **Socio-economic factors:**
 - **Poverty and Unemployment:** Low cost of illicit liquor becomes attractive for poor.
 - **High taxes and regulations** create a demand for cheaper alternative.
 - **Lack of awareness and education** about the harmful impacts
 - **Easier availability:** As it may get produced in the near locality. Further, in states where prohibition is in force, illegality manufactured alcohol become very attractive.
 - **Stronger Impact:** The alcohol content in illicit liquor tends to be very high.
 - » **Legal administrative issues:**
 - **Poor Law and Order situation** due to Police-Criminal Nexus results into limited conviction.
 - **Lack of resources** with authorities make it difficult to effectively monitor and prevent illegal production.
- **Treatment:**
 - Intravenously administer Fomepizole and ethanol. However, fomepizole can be expensive and unavailable in many parts of India. In such scenarios, doctors administer a mixture of ethanol and water (1:1 ratio). Ethanol inhibits methanol's conversion into toxins and help in flushing out of the body either naturally or through dialysis.
- **Way Forward:**
 - » **Make authorities accountable:**

- **The findings of the Mehta Commission** in Gujarat, formed in the aftermath of the 2009 Ahmedabad disaster indicated an **extraordinarily low percentage of bootlegger convictions** and the police's inability to monitor their illicit operations despite having pertinent information.
 - **Stringent policies to check the sale of illicit liquor**
 - » **More regulation of methanol** to prevent its easy availability for illegal liquor manufacturers.
 - » **Encourage community reporting:** If government can develop a mechanism to anonymously report the manufacture and sale of Hooch.
 - » **Developing a balanced policy to drinking alcohol:** Which is based on health science, government's finances and people's right to choose.
 - » **Awareness Generation:** To ensure that vulnerable section is aware of the harmful impacts of liquor in general and spurious liquor in particular.
- **Conclusion:**
- » The war against illicit liquor needs to be fought on multiple fronts and by all stakeholders - Government, NGOs and the community.

3. SPACE – CA UPDATES

1) CHANG'E-6 MISSION: CHINA'S LUNAR PROBE RETURNS TO EARTH WITH WORLD'S FIRST SAMPLES FROM THE FAR SIDE OF THE MOON (JUNE 2024)

- **Example Question:**
 - Discuss the significance of sample return missions from the lunar far side, with specific reference to China's Chang'e-6 [10 marks, 150 words]
- **Introduction:**
 - **Chang' e-6** is a lunar mission by China which landed on the far side of the moon and became the first mission globally to bring back rock and soil samples from the far side of the moon.
 - » The sample has been collected from one of the oldest and largest lunar craters – South Pole Aitken (SPA) basin.
- **What can the lunar samples brought by Chang'e-6 reveal?**
 - **Understanding geographic differences on the moon's two side**. The far side of the moon is geologically different from the near side. It has thicker crust, more craters, and fewer planes where lava once flowed.
 - **Understanding Moon's history and origin**: The collisions that created the SPA basin may have excavated enough material from the Moon's lower crust and upper mantle, which would give insight into the **Moon's history, and possibly its origins**.
 - **Understanding the resources on Moon** which can be extracted or used in future lunar programs.
 - 1) **For e.g.** soil could be used to make bricks to build future lunar research bases through 3D printing.
 - 2) **Similarly**, potential presence of **ice at the moon's pole can** be used to extract hydrogen which can be used in rocket propellant.
 - 3) **Far side** is expected to be rich in **Helium-3** which can become a future source of nuclear energy.
- **Other significance of this sample return mission:**
 - **First samples from far side** – In past, USA, Russia, and China have brought samples only from near side. This is thus an important technological feat.
 - **More detailed analysis of samples using advanced heavy instruments** on earth.
 - **Storage of samples** in future and more detailed analysis when technology advances.
- **Other significance of Chang'e-6 for China:**
 - **Reflects** broader space ambitions and geopolitical aspirations of China.
 - **Enhances international cooperation**: The mission carried payloads from ESA, France, Italy, Pak etc.
- **India's plan of Sample Return Mission**: India is working on Chandryaan-4, and it will be a sample return mission.

Conclusion: The resources on the moon have contributed to a new race to moon. Both USA and China plan to put astronauts on moon before the end of this decade. The success of Chang'e-6 is seen as an important step towards achieving this goal by China.

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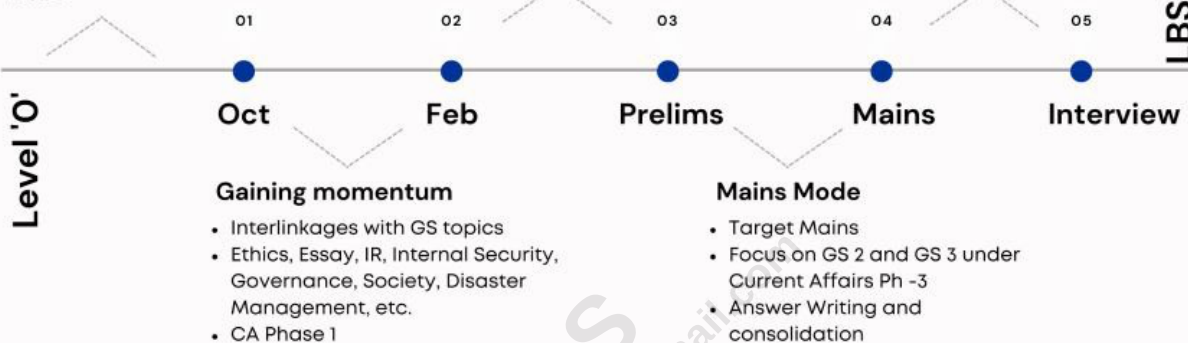
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